Describe a wave

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify amplitude, wavelength, frequency and period.

SUCCESS CRITERIA - I can...

1. Measure amplitude from equilibrium.
2. Measure wavelength between matching points.
3. Connect frequency and period.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “amplitude” and “wavelength” to explain your answer. Sentence starters are optional.

amplitude: maximum displacement from equilibrium for the wave

wavelength: distance between corresponding points on repeated waves

frequency: number of cycles each second

01 Warm-up: bring back an earlier idea

A **3 kg** sample with c=**500 J**/(kg·°C) warms **5°C**. **Find absorbed energy**. If it is **5%** of source energy, find source energy.

Word help: absorbed = taken into the specified object or system

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: f = cycles/Δt; T = **1**/f.
f is frequency (Hz); T is period (s). λ is crest-to-crest wavelength; amplitude is maximum displacement.

Worked reasoning: A wave makes **6 cycles** in **3 s**: f=**2 Hz** and period T=**0.5 s**. Amplitude measures maximum displacement, not the spacing of crests.

Your turn: Draw one spatial wave and **label amplitude** and wavelength **0**. Separately mark one period on a displacement-time graph.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

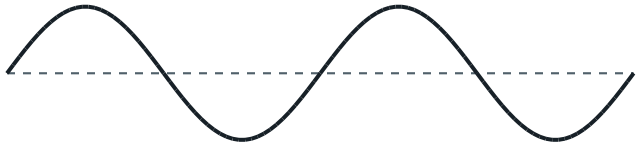


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Label amplitude** and one wavelength on the wave diagram.
2. **Explain why crest-to-trough height** is not the amplitude.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A point on a vibrating string completes **15 cycles** in **0.60 s**. Its crest-to-trough displacement is **0.08 m**.

1. **Find frequency**, period and amplitude.
2. **Sketch a displacement-time trace** and **label one period** and one amplitude.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

Describe a wave

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Does increasing amplitude necessarily increase frequency? Use the definitions to explain **0**.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **0.4 kg** sample absorbs **2400 J** while warming **6°C**. **Find specific heat** and **name one thing** that must be true for this calculation to work.
Label temperatures, samples and relevant energy transfers.

Word help: specific heat = energy per mass needed for a one-degree temperature change

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Sample A

Sample B

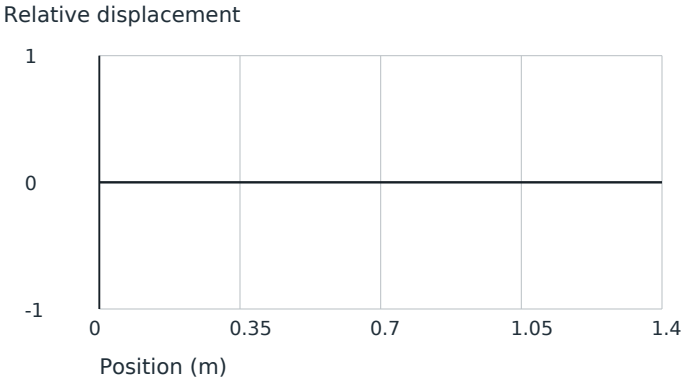
Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

A sinusoidal snapshot has crests at x=**0.20**,**0.70**, **1.20 m**. Find λ. If T=**0.10 s**, find f.

Sentence starter: I found ... [unit]. This value means ... in this situation.

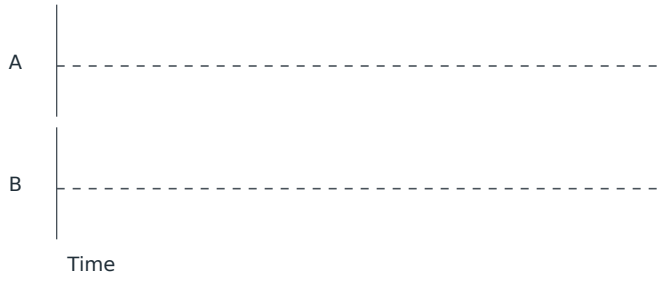


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A point completes **15 cycles** in **6 s**. Successive crests are **1.2 m** apart. **Find period**, frequency and wave speed.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3